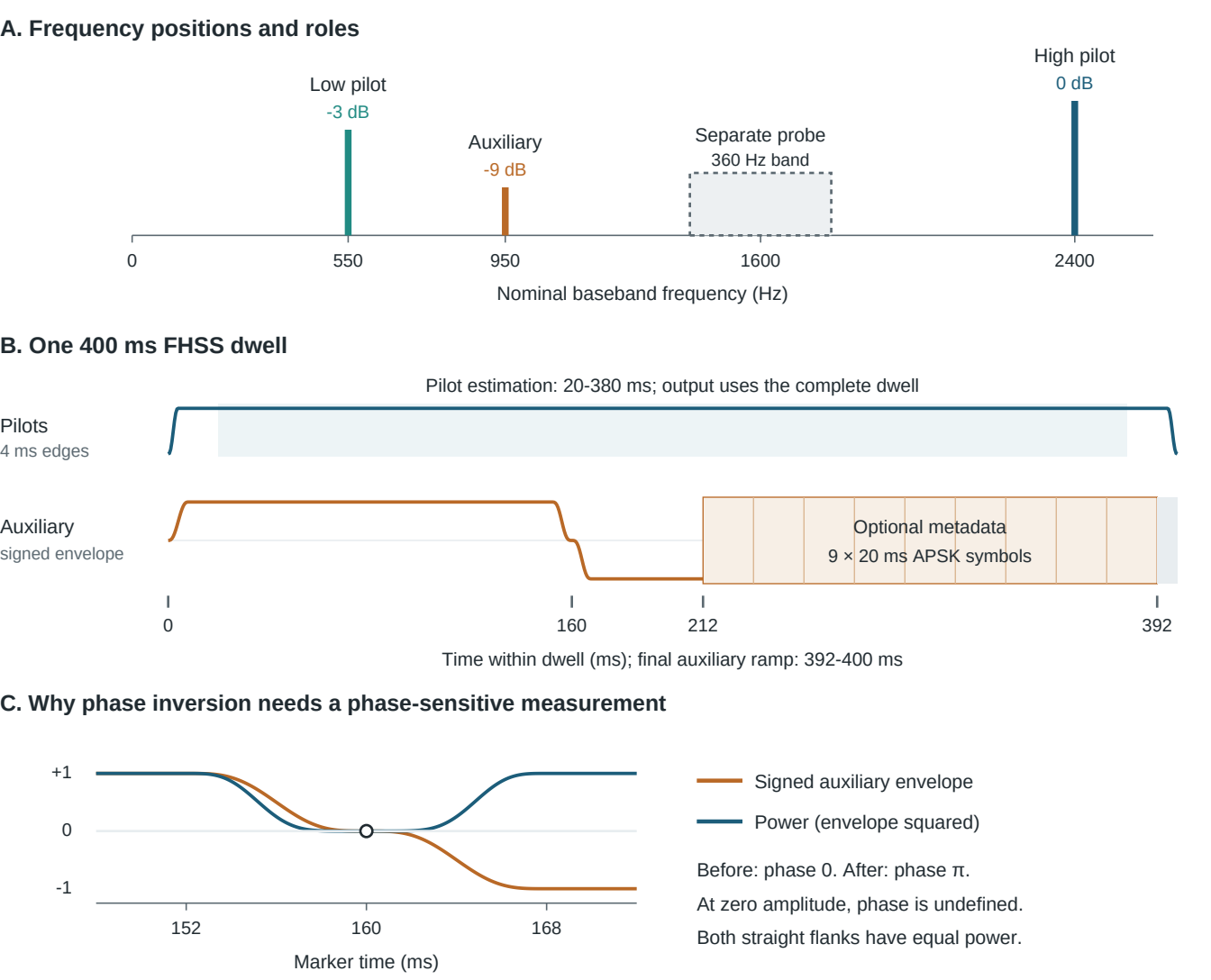
A. Frequency positions and roles
Low pilot
-3 dB
Auxiliary
-9 dB
Separate probe
360 Hz band
High pilot
0 dB
0
550
950
1600
2400
Nominal baseband frequency (Hz)
B. One 400 ms FHSS dwell
Pilot estimation: 20-380 ms; output uses the complete dwell
Pilots
4 ms edges
Auxiliary
signed envelope
Optional metadata
9 × 20 ms APSK symbols
0
160
212
392
Time within dwell (ms); final auxiliary ramp: 392-400 ms
C. Why phase inversion needs a phase-sensitive measurement
+1
0
-1
152
160
168
Marker time (ms)
Signed auxiliary envelope
Power (envelope squared)
Before: phase 0. After: phase π.
At zero amplitude, phase is undefined.
Both straight flanks have equal power.